

Replacement introduction context

Probing Viterbo's Conjecture

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1.1 What remains after the counterexample

Viterbo's conjecture proposed an isoperimetric principle for symplectic capacities: if $K \subset \mathbb{R}^{2n}$ is convex, then

$$c(K)^n \leq n! \operatorname{vol}_{2n}(K),$$

with the ball as the expected equality model Viterbo (2000). For the Ekeland–Hofer–Zehnder capacity in dimension four this becomes

$$\operatorname{sys}(K) := \frac{c_{\text{EHZ}}(K)^2}{2 \operatorname{vol}_4(K)} \leq 1.$$

The ratio removes scale and is unchanged by translations and linear symplectic maps. It therefore asks about shape rather than size.

Haim–Kislev and Ostrover disproved the universal inequality. Their four-dimensional example is the Lagrangian product of two regular pentagons in a suitable relative position, and it satisfies

$$\operatorname{sys}(K_{\text{HKO}}) = \frac{3 + \sqrt{5}}{5} > 1$$

(Haim-Kislev and Ostrover, 2026, Theorem 1.3 and Proposition 1.4). The failure of the proposed bound changes the question rather than ending it. One can still ask how large the ratio can be, which geometric classes obey the inequality, and whether the known counterexample is stable under nearby changes of shape. The centrally symmetric problem is especially important: on products of a centrally symmetric body and its polar, the capacity–volume inequality is tied to Mahler's conjecture Artstein-Avidan et al. (2014).

There is also a precise embedding meaning to the capacity computed in this thesis. Abbondandolo, Edtmair and Kang proved that on every convex body in $\mathbb{R}^4$,

$$c_{\text{EHZ}} = c_Z,$$

where c_Z is the cylindrical capacity (Abbondandolo et al., 2024, Corollary 1). Thus an EHZ computation for one of the four-dimensional convex bodies considered here also determines the smallest symplectic cylinder into which it embeds. The scope of this corollary matters. Their stronger dynamical theorem, asserting that every least-action characteristic bounds a disk-like global surface of section, assumes uniform convexity; the equality of capacities extends to all convex bodies. Neither result identifies the Gromov width with EHZ. Indeed, the HKO counterexample separates the Gromov width from EHZ, while EHZ and the cylindrical capacity still agree.

1.2 The product boundary and the local question

The HKO example belongs to the special class $K \times Q \subset \mathbb{R}_q^2 \times \mathbb{R}_p^2$ of Lagrangian products. In that class the EHZ capacity can be read as the length of a shortest generalized Minkowski billiard trajectory. This connection has recently produced sharp positive results close to the combinatorics explored in this thesis. Balitskiy, Mitrofanov and Polyanskii prove the Viterbo inequality when one planar factor is an arbitrary quadrilateral, and also treat triangular factors and affinely regular hexagonal factors (Balitskiy et al., 2026, Theorem 1.2 and Sections 4–5). Because capacity is symmetric in the two factors, the conclusion applies with either factor in that role.

These results sharply delimit a product search. A triangular or quadrilateral factor cannot yield a new violation while the body remains a Lagrangian product. They do not exclude a general four-dimensional polytope, nor a rotation that mixes the q - and p -planes and destroys the product structure. The regular-pentagon case lies just beyond the quadrilateral theorem and is the setting where the known violation occurs. This makes pentagonal products a natural boundary family, not merely one more low-facet test case.

The thesis studies a different but complementary boundary: perturbations of the counterexample itself. Its main theorem says that K_{HKO} is a local maximum of the systolic ratio among nearby polytopes with the same ten labelled facets, after quotienting by translations, positive dilations and linear symplectic transformations. This is a fixed-facet statement. It does not cover perturbations in which facets appear or disappear, and it does not assert local maximality among all nearby convex bodies.

There is a smooth precedent for asking such a local question. Zoll contact forms, for which all Reeb trajectories are closed with the same minimal period, locally maximize the contact systolic ratio (Abbondandolo and Benedetti, 2023, Theorem 1). The HKO body has corners and a ratio strictly larger than the ball's, so that theorem does not settle its behavior; it instead explains why the organization of minimizing characteristics is relevant to local capacity–volume extremality.

That distinction is also visible in recent work on nonsmooth Zoll-type dynamics. Haim–Kislev studies how EHZ capacity behaves when a convex body is cut by a hyperplane Haim–Kislev (2025). For the HKO pentagon product, Proposition 1.13 proves additivity for specified cut directions near the boundary, while Example 1.12 reports broader numerical evidence and leaves its generalized-Zoll status open. Hyperplane cuts change the facet structure; the theorem of this thesis instead varies the existing ten-facet presentation. The two viewpoints therefore concern adjacent but non-equivalent notions of local rigidity.

1.3 Why finite capacity methods are central here

On a smooth convex boundary, c_{EHZ} is the least action of a closed characteristic. On a polytope the analogous generalized characteristics meet faces and corners, but the relevant optimization can still be made finite. Haim–Kislev expresses the EHZ capacity of a convex polytope as an optimization over ordered facet data and nonnegative weights (Haim–Kislev, 2019, Theorem 1.1). This quadratic-program description is the common input to the thesis's exact capacity computations, its product-specific reductions and the upper functions used in the HKO local certificate.

A second route comes from combinatorial Reeb dynamics. Chaidez and Hutchings associate combinatorial orbits of a four-dimensional convex polytope with Reeb orbits on smoothings

and use the correspondence for computer-assisted searches Chaidez and Hutchings (2021). The flow method in this thesis adapts that geometric viewpoint to an exact finite search under explicit genericity and input hypotheses. The two approaches answer different finite questions: the quadratic program compares global action values, whereas the flow construction follows passages between facets. Their agreement on shared inputs is consequently a substantive cross-check, not a second notation for the same calculation.

These methods support two further parts of the thesis. First, for $P_5 \times R(\theta)P_5$ they yield the complete analytic profile

$$\text{sys}(P_5 \times R(\theta)P_5) = \frac{5 + 2\sqrt{5}}{10} \sec^2 d(\theta),$$

where $d(\theta)$ is the distance to the nearest multiple of $\pi/5$. The HKO position is the maximum within this one-parameter family. This converts an isolated counterexample into a controlled geometric deformation, without claiming a classification of general pentagon products. Second, the exploratory data-science work uses computed capacities to look for reproducible patterns and then asks whether those patterns admit geometric explanations. Its negative searches and subthreshold relations are evidence about the sampled families, not new universal inequalities. Together, the fixed-facet theorem, the finite algorithms, the rotation profile and the exploratory search address the same post-counterexample problem at different scales: which features of a convex body make its shortest symplectic trajectory large relative to volume?

References

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